

HENDERSON-OXFORD, NC, USA

WMO: 720288

Lat: 36.361N

Lon: 78.529W

Elev: 527

StdP: 14.42

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03711

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.4	22.6	-2.2	5.0	23.8	3.2	6.6	28.7	20.0	57.4	18.2	51.4	4.1	310	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	93.1	73.6	90.4	72.7	87.9	72.1	76.3	86.4	75.3	85.1	74.3	83.5	5.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.3	126.4	78.8	72.7	123.5	78.3	72.0	120.9	77.9	40.0	87.9	39.0	84.9	38.2	83.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.1	15.0	12.5	DB	11.8	97.0	5.7	3.5	7.7	99.6	4.4	101.6	1.2	103.6	-2.9	106.2	(4)
(5)			WB	10.1	78.0	5.2	1.4	6.3	79.0	3.3	79.8	0.3	80.5	-3.4	81.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.1	39.7	42.5	49.3	58.9	66.5	74.1	77.6	76.1	71.1	59.9	49.0	43.7	(6)	
	DBStd	15.48	10.42	9.58	10.02	7.92	7.08	5.39	4.34	4.50	5.58	8.10	8.60	9.23	(7)	
	HDD50	1129	348	248	141	15	0	0	0	0	0	14	120	243	(8)	
	HDD65	3557	783	631	492	213	72	4	0	1	16	201	482	662	(9)	
	CDD50	4459	29	39	120	282	513	722	856	809	631	319	91	48	(10)	
	CDD65	1412	0	1	6	30	120	276	391	345	197	41	3	2	(11)	
	CDH74	11823	0	2	55	287	942	2445	3542	2841	1407	284	13	5	(12)	
	CDH80	4224	0	0	3	40	245	905	1468	1069	442	52	0	0	(13)	
(14) Wind	WSAvg	5.1	5.9	6.0	6.3	6.7	5.4	4.6	4.3	3.6	4.1	4.6	4.9	5.1	(14)	
(15) Precipitation	PrecAvg	44.3	3.6	3.1	4.1	3.1	3.7	3.8	4.3	4.9	4.0	3.3	3.2	3.2	(15)	
	PrecMax	62.7	7.3	5.8	8.7	6.0	7.8	10.0	13.5	12.0	17.1	9.7	10.0	6.4	(16)	
	PrecMin	31.2	0.7	0.2	0.7	0.6	0.7	0.4	1.0	1.0	0.5	0.0	0.5	0.3	(17)	
	PrecStd	7.7	1.7	1.5	1.9	1.6	1.7	2.5	2.2	2.7	3.1	2.3	1.9	1.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.0	73.1	79.4	84.3	89.4	95.4	97.8	94.9	91.0	85.4	76.6	72.7	(19)	
		MCWB	60.6	59.8	61.2	62.6	70.5	72.7	73.3	73.7	70.1	68.8	63.2	64.2	(20)	
	2%	DB	64.3	68.4	74.5	80.8	85.9	91.2	93.5	91.3	87.8	80.7	71.5	67.5	(21)	
		MCWB	57.7	56.8	59.4	62.9	68.4	72.3	74.1	73.8	70.7	67.0	59.9	62.1	(22)	
	5%	DB	61.4	63.8	71.1	77.4	82.5	88.2	90.6	88.4	85.4	77.1	67.7	63.5	(23)	
		MCWB	55.6	54.4	57.2	62.1	67.7	71.8	73.3	72.8	70.0	65.4	59.4	58.6	(24)	
	10%	DB	56.8	59.2	65.9	73.4	79.7	85.7	87.8	86.0	82.1	73.1	63.7	59.2	(25)	
		MCWB	49.6	50.4	55.0	60.6	66.1	70.7	73.0	71.9	69.0	65.1	57.2	53.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.4	62.8	64.5	69.4	73.2	76.3	77.4	78.0	75.8	73.3	67.0	66.9	(27)	
		MCDB	66.9	69.3	73.4	75.7	83.5	86.5	88.7	88.2	82.9	76.4	71.3	70.0	(28)	
	2%	WB	60.0	60.1	62.3	67.0	71.1	74.9	76.3	76.4	74.0	71.6	64.1	63.1	(29)	
		MCDB	63.9	65.7	70.2	74.6	80.5	85.3	87.2	86.0	81.0	75.8	67.8	65.8	(30)	
	5%	WB	56.8	56.7	60.2	64.8	69.6	73.8	75.3	75.2	72.9	69.6	61.4	59.6	(31)	
		MCDB	61.3	63.2	67.6	72.3	78.4	84.4	85.9	84.3	79.5	74.6	66.0	62.4	(32)	
	10%	WB	51.4	51.9	57.6	62.8	68.4	72.7	74.4	74.0	71.5	66.5	58.3	54.5	(33)	
		MCDB	55.9	57.4	64.1	70.6	76.4	82.5	84.1	82.2	77.8	72.4	63.4	57.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	17.7	18.9	20.5	21.6	19.7	19.1	18.6	17.8	18.3	19.5	19.5	17.2	(35)	
		MCDBR	22.4	24.4	24.8	25.3	22.4	21.8	22.0	21.0	22.1	22.7	23.0	20.1	(36)	
	5% WB	MCWBR	16.7	15.7	13.2	11.6	9.0	7.0	6.5	6.8	7.9	11.2	14.1	15.2	(37)	
		MCWBR	20.3	21.8	21.4	19.9	18.9	19.3	18.5	17.9	17.3	17.6	18.9	18.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.304	0.321	0.352	0.395	0.443	0.488	0.523	0.522	0.409	0.366	0.323	0.307	(40)	
		taud	2.517	2.467	2.419	2.300	2.193	2.113	2.033	2.017	2.331	2.438	2.516	2.545	(41)	
	Ebn at Noon		284	292	290	281	268	255	245	242	268	271	274	276	(42)	
		Edh at Noon	25	30	34	41	46	50	54	54	37	30	25	23	(43)	
(44) All-Sky Solar Radiation	RadAvg		793	1017	1335	1690	1875	2032	1908	1728	1422	1149	875	689	(44)	
	RadStd		57	117	98	133	148	139	108	89	128	143	71	60	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (58 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+283	+187

Nomenclature: See separate page